Part: **3.5 – SANDWICH PANELS**

Chapter: **3510**

Title: **STRENGTH OF COMPOSITE SANDWICH PANELS**

Revision: **1.2**

Date: **15/03/2022**

1 INTRODUCTION

This chapter defines the methodology that shall be used to perform the (local) strength check of honeycomb sandwich panels.

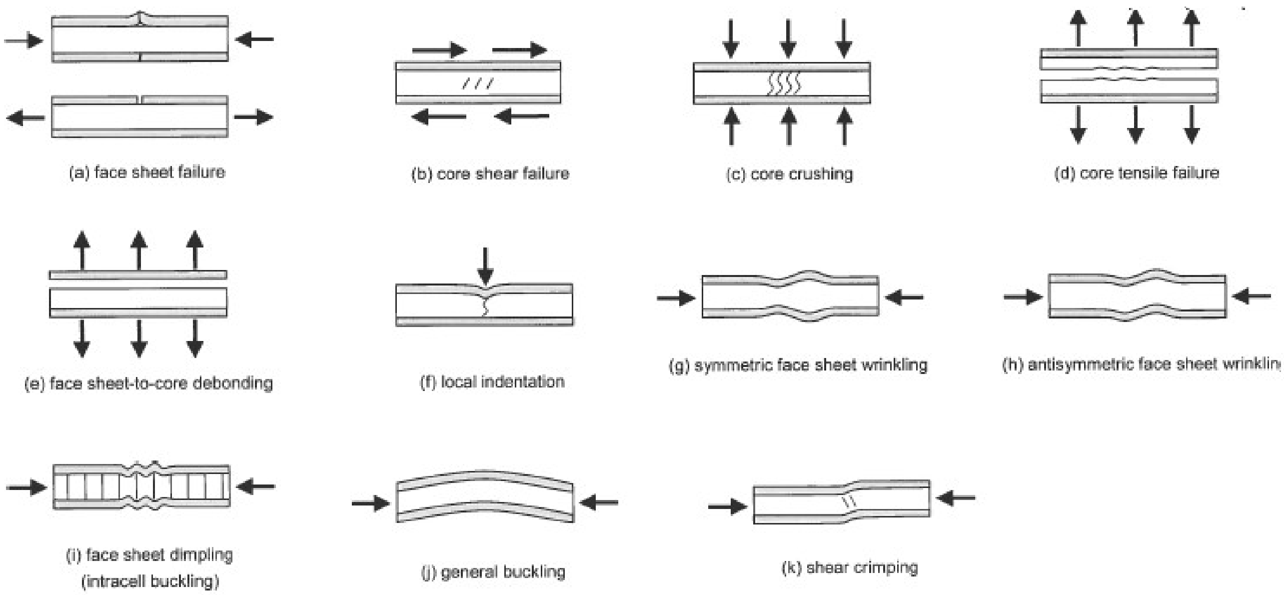


Figure 1. Typical failure modes of sandwich panels, extract from CMH-17 Vol. 6

The calculation of the internal loads distribution within the sandwich panel, as well as the global buckling analysis, is out of the scope of this chapter. See previous introductory chapter 3500 “SANDWICH PANELS” and 3530 “RECTANGULAR SANDWICH PANELS” for advice on these topics.

2 METHOD DESCRIPTION

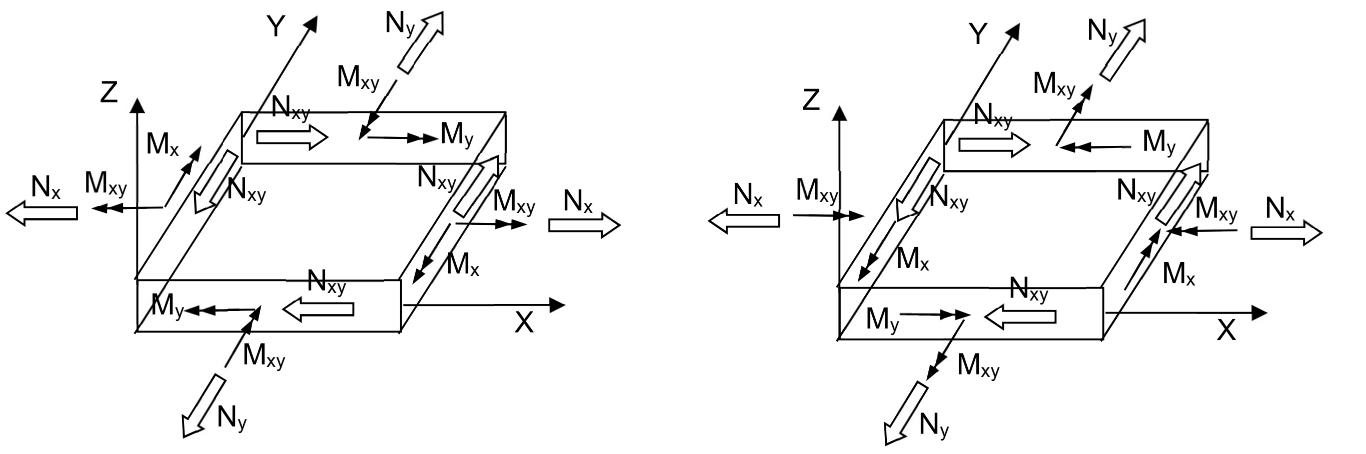
2.1 STRESSES AND STRAINS IN CORE AND FACES

Methodologies for calculating the internal loads distribution in a sandwich panel, typically in terms of plate loads (force and moment flows), is out of the scope of this chapter. Internal loads at each location shall be available by other means, typically FEM analysis.

Stresses and strains within the sandwich materials shall be computed from plate/shell loads using the Classical Laminate Theory.

- ✓ The sandwich-laminate stacking in this analysis includes the lower face plies, the core-ply (with 0 membrane and bending stiffnesses) and the upper face plies. The standard convention is that ply stacking starts (ply 1) with lower face external ply at z = -t/2 and ends (ply N) at upper face external ply at z = +t/2.

- ✓ Sometimes the internal loads in the sandwich panel are computed using Finite Elements Analysis (FEA) with MSC-Nastran, modelling the sandwich panel with 2D shell elements only (only one shell element through the thickness). In such cases, internal loads will be available in terms of MSC-Nastran element forces, with a particular sign convention. Even if MSC-Nastran and Bibliography reference axis systems were equivalent, it will be necessary to make a conversion, changing M_x, M_y, and M_{xy} signs, to make the analysis consistent.



NASTRAN FORCES (M>0 compression at z>0) BIBLIOGRAPHY FORCES (M>0 tension at z>0)

Figure 1. Signs criterion for element forces

If the sandwich panel is not modelled with 2D elements, but with 3D elements (of height *H*) for the core, and 2D elements for the faces, the equivalent plate loads would be:

$N_x = (N_{x\text{-}lo} + N_{x\text{-}up}) / 2$

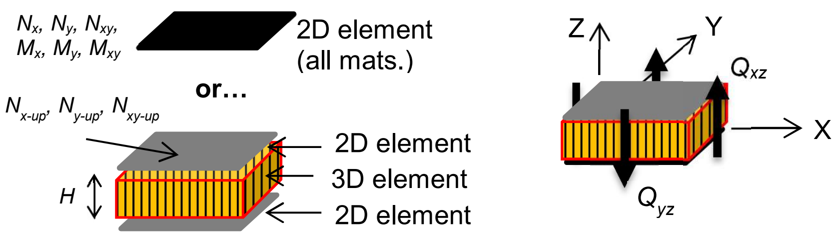
$N_y = (N_{y\text{-}lo} + N_{y\text{-}up}) / 2$

$N_{xy} = (N_{xy\text{-}lo} + N_{xy\text{-}up}) / 2$

$M_x = (N_{x\text{-}up} - N_{x\text{-}lo}) \cdot H$

$M_y = (N_{y\text{-}up} - N_{y\text{-}lo}) \cdot H$

$M_{xy} = (N_{xy\text{-}up} - N_{xy\text{-}lo}) \cdot H$



The same relationships could be used to obtain the faces (in-plane) forces from the membrane and bending forces in the sandwich plate assembly (in this case the sum of the core height plus faces semi-thickness should be included in *H*).

Transverse shear loads (forces per unit length) within the sandwich plate are Q_{xz} and Q_{yz} .

2.1.1 MEAN/APPARENT STRESS IN FACES

Many of the methods proposed to check the local failure modes of the sandwich panel are referenced to the mean, or apparent, stresses at the faces. It is remarked that these stresses are not the ply stresses.

If in-plane loads (forces per unit length) at face mid-line are available (typically noted as F_x, F_y, F_{xy} , or N_x, N_y, N_{xy}) these mean/apparent stresses can be obtained simply by dividing those forces by the face thickness.

It is also possible to obtain these values from face mid-line strain and the apparent elastic moduli (or membrane stiffness matrix) of the face laminate.

2.1.2 APPARENT ELASTIC MODULI

For some of the methods to check the sandwich strength either the elastic moduli of the faces, or the transverse shear of the core, must be evaluated in an off-axis direction relative to the local reference systems (that based on the face 0º material, or that based on the core ribbon direction).

- ✓ Core shear modulus at any transverse shear plane can be calculated in the loading direction with formula in CMH-17 - Vol. 6, chapter §4.6.2 as:

$$G_\theta = \frac{G_L G_W}{G_L \sin^2\theta + G_W \cos^2\theta}$$

where G_L, G_W and G_θ are the shear modulus in the ribbon, transverse and loading direction respectively.

- ✓ The apparent elastic moduli of the face at any direction, assuming the laminate is quasi-orthotropic, can be obtained using expressions in Jones (Ref. 2, formulas 2.97)

Remark: even face laminate were not symmetric, unless otherwise specified, for the rest of analyses in this chapter the apparent elastic moduli of the faces shall be calculated as if the face-laminate was indeed symmetric. However, and especially in cases with non-symmetric laminates in faces and/or faces affected by waviness/telegraphing defects, it should be mandatory that design values of the part are confirmed with dedicated tests representative of actual configuration (typically 4-points-bending and/or edgewise-compression tests).

2.2 LOCAL STRENGTH OF HONEYCOMB SANDWICH PANELS

Most of the proposed strength-check methodologies of the local failure modes of the sandwich panels are those recommended by the CMH-17 - Vol. 6 (Ref. 1) in its chapter §4.6.

2.2.1 PLAIN STRENGTH OF FACES

Use the method described in chapter 3411 "PLAIN STRENGTH OF COMPOSITE LAMINATES" to perform the plain strength check of the sandwich faces.

As for solid laminates, sandwich face laminates shall include plies in a sufficient number of directions capable to sustain a general plane stress status with axial stresses/strains in reinforcement fibers (in more than 3 directions, typically 0º, 90º, 45º and -45º). On the other hand, faces laminates are not expected to be symmetric per se. Even though, failure criteria applicable to symmetric solid laminates shall be still applicable (the bonding to the core prevents the local membrane-coupling effects). Face forces (see previous section), or strain level at faces mid-line from classical laminate analysis on the whole sandwich assembly shall be used.

Anyway, certain production processes (co-curing or co-bonding) introduce waviness in the faces. Dedicated test should support real strengths in those cases.

EN SANDRES parece ser que a los admisibles luego le meten un factor Kb de penalización cuando se tratan de calculos de pandeo.

¿Es el mismo que el Kb B-basis?

2.2.2 DIMPLING

Face sheets may buckle or dimple into the spaces between core cell walls. The sandwich strength vs this failure mode is check using the method proposed in CMH-17 - Vol. 6, chapters §4.6.5.1(c) (strength in compression), §4.6.5.3 (strength in shear), and §4.6.5.4 (failure law and reserve factor).

Applied internal loads are the faces in-plane apparent stresses σ_x , σ_y , and τ_{xy} .

- ✓ Compression strength in dimpling can be obtained from the face thickness (*t*), the cell size (*s*) and the terms of the bending stiffness matrix of the face as:

$$F_c = (1/t)\cdot(\pi/s)^2\cdot[D_{11} + 2\cdot(D_{12} + 2\cdot D_{66}) + D_{22}]$$

  If face laminate were not symmetric, the generalized bending stiffness matrix *[D*]* shall be used instead (see CMH-17 - Vol. 3, eq. 9.2.2.1(b)), i.e. the stiffness referenced to the plane where no membrane-bending coupling occurs, being $[D^*] = [D] – [B] \cdot [A]^{-1} \cdot [B]$

- ✓ Shear strength in dimpling can be obtained from the face apparent Young modulus of the face as:

Pone que solo para cuando es no simetrico pero hay que mirar el CMH-Vol3 porque creo que tambien es para cuando no es balanceado, porque creo que en ese caso la matrix B es distinto a la nula

$$F_s = 0.6 \cdot \min(E_x, E_y) \cdot (t/s)^{1}$$

Interaction between failure indices in axial stress shall be linear. Proposed interaction between failure indices in compression and shear is the typical criterion for buckling, i.e. failure at $R_c+R_s^2=1$.

$$R_c = \max(0, -(\sigma_x+\sigma_y)/F_c) \qquad R_s = |\tau_{xy} / F_{xy}| \qquad RF = 2 / [R_c + (R_c^2 + 4\cdot R_s^2)^{0.5}]$$

2.2.3 WRINKLING

Wrinkling failures may occur if a sandwich face sheet buckles as a plate on an elastic foundation. The sandwich strength vs this failure mode is check using the method proposed in CMH-17 - Vol. 6, chapters §4.6.6.3(a)(b) (allowable strength), §4.6.6.5 , Ref. 4 and Ref. 6 - §10.3.3 (failure law).

Applied internal loads are the faces principal stresses σ_1 and σ_2 , obtained from in-plane apparent stresses σ_x , σ_y , and τ_{xy} , using the classical formulation (Mohr's circle).

Allowable compression stress in each principal direction is computed using the following expressions, from face apparent elastic modulus in applicable direction (E_f), core elastic modulus in through-thickness direction (E_c), transverse shear modulus of the core in applicable direction (G_c), the core thickness (t_c) and the face thickness (t)…

En SANDRES parece que para un core con G13 distinto a G23 y laminados iguales y quasi-iso, sale admisibles iguales. Lo cual reproduciendolo en mis calculos no salen igual

- ✓ For "thick" cores: $F_w = C_1 \cdot (E_f \cdot E_c \cdot G_c)^{1/3} + C_2 \cdot G_c \cdot t_c / t_f$
- ✓ For "thin" cores: $F_w = C_3 \cdot [E_f \cdot E_c \cdot (t_f / t_c)]^{0.5} + C_4 \cdot G_c \cdot t_c / t_f$

…where "thick" cores are those for which:

$$t_c = 1.82 \cdot t_f \cdot (E_f \cdot E_c / G_c^2)^{1/3}$$

and constants C_1, C_2, C_3, C_4 are configuration specific and may be adjusted from tests. Default conservative recommended values are:

$$C_1 = 0.247,\ C_2 = 0.078,\ C_3 = 0.33,\ C_4 = 0$$

Two approaches are mentioned in that reference for obtaining the RF for bi-axial stress states. Plantema suggests to use the minimum of the two margins (uncoupled) for each of the principal stresses σ_1 and σ_2. Wart and Gintert proposes a more complicated law, with different exponents function of the comparison between σ_1 and σ_2 values. Not accounting for any interactions seems coarse and possibly not conservative, and the second option would present discontinuities when the stresses in both directions were similar but the allowables different, i.e. when $\sigma_1 \approx \sigma_2$ and $F_{w1} \neq F_{w2}$.

the following interaction equation. The following equation is used when f_{yf} is the maximum compressive stress. When f_{xf} is the maximum compressive stress, the y-term should be cubed instead of the x-term. If only one load is compressive, the cubed exponent is left off:

$$MS_{Cwrinkle} = \frac{1}{\left(\frac{f_{xf}}{F_{xfwrinkle}}\right)^3 + \left(\frac{f_{yf}}{F_{yfwrinkle}}\right)} - 1$$

Figure 2. Extract of Ref. 4 with the interaction formula for biaxial compression in wrinkling

A slightly modified version of the Wart and Gintert criterion is proposed as design criterion:

- ✓ When σ_1 and σ_2 are both compressive stresses:
  - if ($|\sigma_1/F_{w1}| > |\sigma_2/F_{w2}|$) $RF = 1 / [(\sigma_1/F_{w1}) + (\sigma_2/F_{w2})^3]$
  - if ($|\sigma_2/F_{w2}| > |\sigma_1/F_{w1}|$) $RF = 1 / [(\sigma_1/F_{w1})^3 + (\sigma_2/F_{w2})]$
- ✓ If stress is tension at any of the directions, that contribution shall be conservatively ignored, i.e. the wrinkling RF obtained for the direction in compression shall be directly used.

2.2.4 SHEAR CRIMPING

Core shear crimping is a general instability failure in which the wave length of the buckles is very small because the core shear modulus is low. The sandwich strength vs this failure mode is check using the method proposed in CMH-17 - Vol. 6 - §4.6.7.

Applied internal loads are the faces in-plane apparent stresses σ_x , σ_y , and τ_{xy} .

Allowable stresses are calculated from faces thicknesses (t_L, t_U), core thickness (t_c), distance between middle surfaces of faces (H) and transverse shear stiffness in each direction (G_{xz}, G_{yz}), as:

$$F_{cx} = H^2 \cdot G_{xz} / [(t_U + t_L) \cdot t_c] \qquad F_{cy} = H^2 \cdot G_{yz} / [(t_U + t_L) \cdot t_c] \qquad F_{xy} = H^2 \cdot (G_{xz} \cdot G_{yz})^{0.5} / [(t_U + t_L) \cdot t_c]$$

In computing failure indices and final reserve factor tension stresses are ignored. Proposed interaction between failure indices in compression and shear is the typical criterion for buckling, i.e. failure at $R_c + R_s^2 = 1$.

$$R_c = max\ (0,\ -\sigma_x / F_{cx},\ -\sigma_y / F_{cy}) \qquad R_s = |\tau_{xy} / F_{xy}| \qquad RF = 2 / [R_c + (R_c^2 + 4 \cdot R_s^2)^{0.5}]$$

2.2.5 CORE SHEAR

Core shear strength shall be checked as proposed by CMH-17 - Vol. 6 - §4.6.2.

The core shear stress is obtained from transverse shear loads (Q_{xz} and Q_{yz}) and core thickness (t_c). If X is the core longitudinal/ribbon direction, and Y the core transverse direction, the value of that stress, in plane θZ (angle θ reference to X axis), is:

$$\tau_\theta = (Q_{yz}^2 + Q_{xz}^2)^{0.5} / t_c \qquad tan(\theta) = Q_{yz} / Q_{xz}$$

The allowable stress in the shear plane, from core shear allowable stresses in ribbon (F_{sL}) and transverse (F_{sW}) shear planes:

$$F_{s\theta} = 1 / \{ [\cos(\theta) / F_{sL}]^N + [\sin(\theta) / F_{sW}]^N \}^{1/N} \qquad \ldots \text{with } N = 1.45$$

The, the reserve factor is simply: $RF = F_{s\theta} / \tau_\theta$

2.2.6 CORE CRUSHING

Core crushing strength shall be checked as proposed by CMH-17 - Vol. 6 - §4.6.3.

Thus, core compression stress in through thickness direction is:

$$f_{crush} = \frac{M_x^2}{d D_x} + \frac{M_y^2}{d D_y} + q$$

f_{crush} = Core compressive stress
M_X, M_y = Bending moments from in-plane and pressure loads
D_X, D_Y = Flexural stiffnesses for beams parallel to the X and Y directions, respectively
d = Distance between face sheet centroids
q = Normal pressure

If core strength in through-thickness compression is F_{3c}, the reserve factor is then: $RF = F_{3c} / f_{crush}$

2.2.7 RESIDUAL STRENGTH AFTER IMPACT

Use the method described in chapter 3413 “RESIDUAL STRENGTH AFTER BVID IMPACT” to perform the residual strength analysis of the sandwich faces, including an undetected impact damage (if this requirement is applicable to the component).

It is noted that some test campaigns on sandwich coupons have shown an appreciable sensitivity of *CAI* residual strength to direction of applied load, even with faces that were quasi-isotropic. During *CAI* tests on sandwich coupons representative of A400M Flap Support Fairings (non-symmetric thin faces t_f<1mm made of CFRP UD-tape and Nomex honeycomb core), the ratio of *CAI* strengths in core transverse direction to longitudinal/ribbon direction was around the ratio that would be obtained for wrinkling strengths i.e. $\sim[G_L/G_W]^{1/3}$ (see Ref. 3). This phenomenon can be only explained by some kind of interaction between failure modes driven by the laminate (*CAI*) and failure modes related to the honeycomb core. Thus, design values for damage tolerance strengths of sandwich faces should be validated by dedicated tests in core ribbon and transverse directions, especially in configurations accentuating differences with standard solid laminates (not-symmetric laminate stacking, thin faces or waviness/telegraphing defects).

2.2.8 OTHER FAILURE MODES

Flatwise core tension failure or faces disbonds are not normally critical, prevented by an adequate selection of materials, production and quality control processes, as well as a proper qualification at those levels.

Analysis of failure modes related to local load introductions (inserts, bolts, …) are out of the scope of this chapter.

2.3 RAMPS

At the edges of most sandwich panels, one face sheet, the bag-side sheet for composite construction, ramps up over the core. Interlaminar stresses are induced at the ends of the ramp which can cause core failure (with through thickness compression) or debonding between core and face sheet (with flatwise tension).

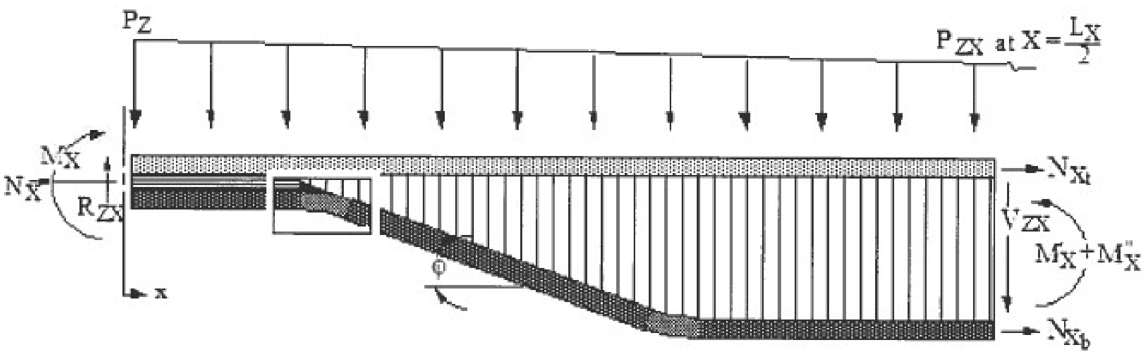


Figure 3. Geometry of sandwich edge, from CMH-17 Vol. 6

The following approach is that proposed by the CMH-17 Vol. 6 sub-section §4.6.3 to compute those interlaminar stresses, and the internal loads in face sheets and core. The following equations use loads in the X direction. The loads in the Y direction must also be checked using similar equations.

$$f_{X\,flat} = \pm \frac{N_{xb}}{R}$$

Use "+" for concave radii and "-" for convex radii
$F_{X\,flat}$ Flatwise stress at radius
N_{xb} Load in the bag-side face sheet
R Ramp radius
ϕ = ramp angle
θ = local angle between bag side and tool side in the ramp region

$$N_{Xt} = \frac{N_X}{2} - \frac{M'_X + M''_X}{d} \qquad N_{Xb} = \left[\frac{N_X}{2} + \frac{M'_X + M''_X}{d}\right]\left(\frac{1}{\cos\theta}\right)$$

$$V_{ZX} = \left[R_{ZX} - P_Z(x) + \left(\frac{P_Z - P_{ZX}}{L_X}\right)\left(x^2\right)\right] - \left[N_{Xb}(\sin\theta)\right]$$

Figure 4. Ramp flatwise stress, face-sheets in-plane loads and transverse shear at core

At the radius closest to the edgeband, the critical stress is considered to be interlaminar tension between the plies of the bag-side laminate.

At the radius further to the edgeband the failure modes are considered to be either separation of the bag-side face sheet from the core (flatwise tension, when the bag-face is loaded in in-plane compression, i.e N_{xb} >0), or core crushing (flatwise compression, when the bag-face is loaded in in-plane tension).

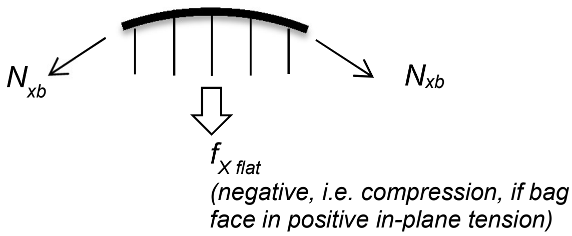


Figure 5. Flatwise compression in inner ramp end with bag sheet in tension (convex radius)

2.4 LOCAL STRENGTH OF FOAM SANDWICH PANELS

If the sandwich panel is constructed with a rigid foam core, the methodology to perform the strength check shall be similar, but taking into account a few specificities:

- ✓ the dimpling failure is not possible (no cells),
- ✓ the wrinkling stress shall be calculated using the dedicated expression in CMH-17 - Vol. 6 - chapter §4.6.6.2 (continuous support)

$$F_W = Q\left(\frac{E' E_C G_C}{\lambda}\right)^{1/3}$$

Q	adjustment parameter, 0.5 default (conservative)
E'	bending elastic modulus of the face in load direction, derived from the stiffness matrix of the face laminate *[D*]*
Ec	core elastic modulus
Gc	core shear modulus
$\lambda = 1 - v^2$	with *v* the Poisson's ratio of the core, default 0.25

and…

✓ the static margin for the core material shall be obtained taking into account the interaction of through-thickness axial and shear stresses by a typical quadratic law, i.e.

$$RF = 1 / (R_z^2 + R_s^2)^{0.5}$$

where: $R_s = \tau_\theta / F_{su\text{-}c} = (\tau_{xz}^2 + \tau_{yz}^2)^{0.5} / F_{su\text{-}c}$ is the failure index of the core in transverse shear

$R_z = \sigma_z / (F_{zt}\ ,\ - F_{zc})$ is the failure index of the core in through-thickness tension, or compression

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
1	-	As subscript, relative to principal stress direction (that with the maximum stress)
2	-	As subscript, relative to principal stress direction (that with the minimum stress)
[A]	N/mm	Stiffness matrix of a composite laminate in in-plane (membrane) loads
[B]	N	Stiffness matrix of a composite laminate determining the membrane/bending coupling
b	-	As subscript, relative to bag face
BVID	-	Barely Visual Impact Damage
c	-	As subscript, relative to the core of the sandwich panel, or indication a compression load/stress (from context)
CAI	MPa / με	Residual strength in compression a BVID (in terms of stress or strain)
CFRP	-	Carbon Fibre Reinforced Polymer
d / H	mm	Distance between face sheet centroids
[D]	Nmm	Stiffness matrix of a composite laminate in out/plane (membrane & torsion) moments
E	MPa	Elastic Young's modulus
F	MPa	Allowable stress
f	MPa, -	Stress or, as subscript, relative to a face of the sandwich panel
G	MPa	Shear modulus

L , l	-	As subscript, relative to lower face of the sandwich panel, or indicating the longitudinal/ribbon direction of the honeycomb core, from context
M	Nmm/mm	Shell out-plane (bending and torsion) moments
N	N/mm	Shell in-plane (membrane) loads
Pz	MPa	Pressure force applied on tool face
Q, V	N/mm	Shell transverse shear load
R	- / mm	Failure index (applied load divided by allowable load), typically in an uni-axial stress state, or radius of a ramp end, from context
R_{ZX}	N/m	Reaction force at sandwich edgeband
RF	-	Reserve Factor
s	mm / -	Honeycomb cell size or, as subscript, indication of a shear load/stress
t	mm / -	Thickness of a face or thickness of the whole sandwich, from context, or as subscript, relative to tool face
U , u	-	As subscript, relative to upper face of the sandwich panel
UD	-	Uni-Directional
X, Y, Z	-	Direction of a local reference axis system, with X and Y parallel to sandwich plane, and X aligned with material reference 0º direction
w	-	As subscript, relative to wrinkling failure
W	-	As subscript, indicating the transverse direction of a honeycomb core
θ	-	Angle between a direction in the sandwich plane and the X reference axis, or between the bag-face at ramp and the tool face, from context
σ	MPa	Applied axial stress
τ	MPa	Applied shear stress

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	[SAE] CMH-17	2013	COMPOSITE MATERIALS HANDBOOK. VOLUME 6. STRUCTURAL SANDWICH COMPOSITES.
Ref. 2	[Robert M. Jones]	2nd Ed.	MECHANICS OF COMPOSITE MATERIALS
Ref. 3	TAE-A4-NT-170001	Is. A	A400M FSF SANDWICH PANELS. ANALYSIS OF CAI TEST RESULTS.
Ref. 4	[S. Ward, L. Gintert]	2001	Analysis of Sandwich Structures
Ref. 5	TAE-A4-MM-170791	Is. A	A400M FSF. MATERIAL DESIGN VALUES AND METHOD FOR POST-CERTIFICATION STRENGTH ANALYSES WITH TELEGRAPHING DEFECTS.
Ref. 6	[C. Kassapoglou]	2nd Ed.	Design and Analysis of Composite Structures

SOFTWARE

Program	***Version***	***Description***
NASTRAN (MSC-Software)	2005	Multidisciplinary structural analysis software based in the Finite Elements methods
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures
PANSA (Airbus DS)	5.5	Stress, strength and buckling analysis of composite sandwich panels
SAM - SANDWICH (Airbus DS)	1.6	Local strength analysis of composite sandwich panels

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
1.0 26/06/2019	G. Baños	First issue	All pages
1.1 16/07/2021	G. Baños	Explain differences in method if core is foam (continuous support) instead of honeycomb	New section 2.4
1.2 28/02/2022	G. Baños	Clarify meaning of height H used in shear crimping and other formulas	Pages 6, 9, 11

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Methods (TEPAP-TL4)	G. Baños 28/02/2022	Florian Glock 13/03/2022

Table 3. *Approval signatures table for last revision of this chapter*